Loyola Marymount University

Safe and Reactive Robotic Arm for Ball Interception and Obstacle Avoidance

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Senior Capstone 2025-2026

LMU Department of Electrical and Computer Engineering

Meet the team

Makari Green

Lead on GPU integration and ball tracking. Responsible for sensor data handling, feature extraction, and trajectory computation.

Arye Mindell

Lead on ROS control and motion planning. Develops motion trajectories, synchronization with ROS, and actuator control logic.

Daniel Childs

Lead on system testing and safety integration. Implements obstacle detection algorithms, safe-abort logic, and test data collection. Assists with ROS implementation and final performance evaluation.

Problem

Safely catch a ball, avoiding static obstacles.

Importance

Robots for dynamic work and life, safely interact in real time

Motivation

Exploring high-speed perception and safe motion control.

Interactive demonstrations for students.

Loyola Marymount University

https://brand.lmu.edu/templatesanddownloads/printstationery/

Ethics and Sustainability

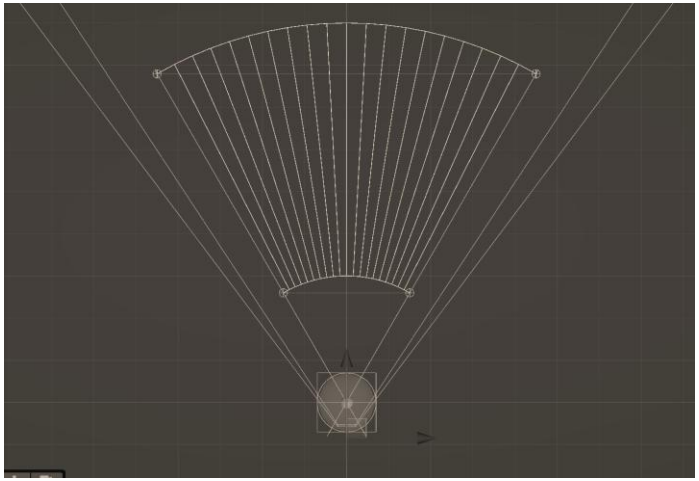
- Focus on making robots a more trustworthy/reliable companion in work and personal life
- Improve human safety in potentially dangerous automated environments

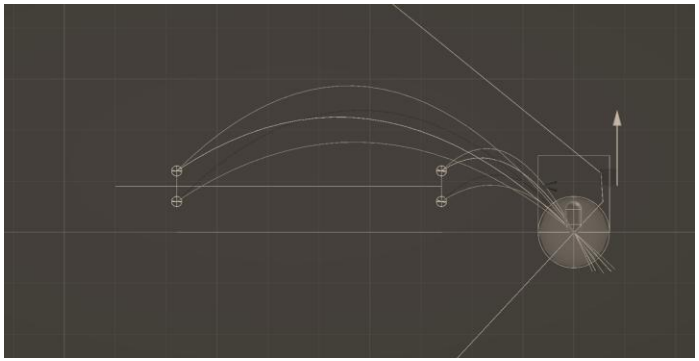
- **SDG 3**: Good Health and Well-Being
  - Safe, effective human-robot interaction, potential enable robots to relieve humans of dangerous tasks
- **SDG 9**: Industry, Innovation, & Infrastructure
  - Robot is reusable in future EE projects, test rig designed for re-use
  - Enable robots to become an effective, reliable contributor to industry

Image Source: https://www.arenasolutions.com/resources/glossary/sustainable-development-goals/
Accessed: October 15, 2025

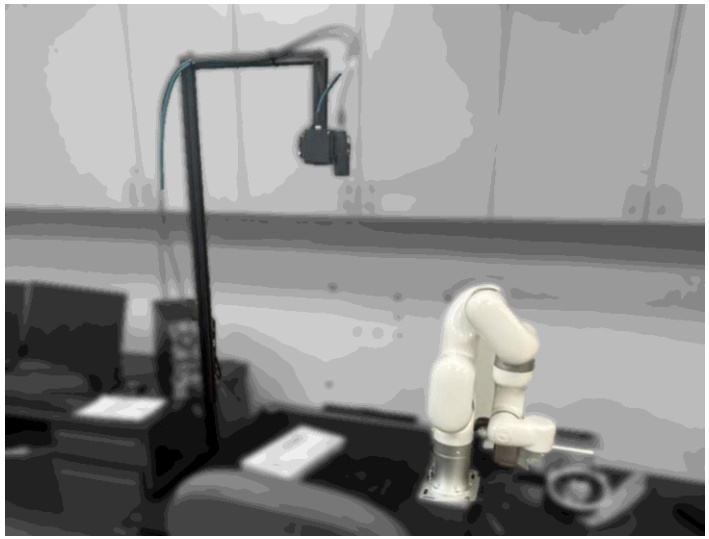
Requirements

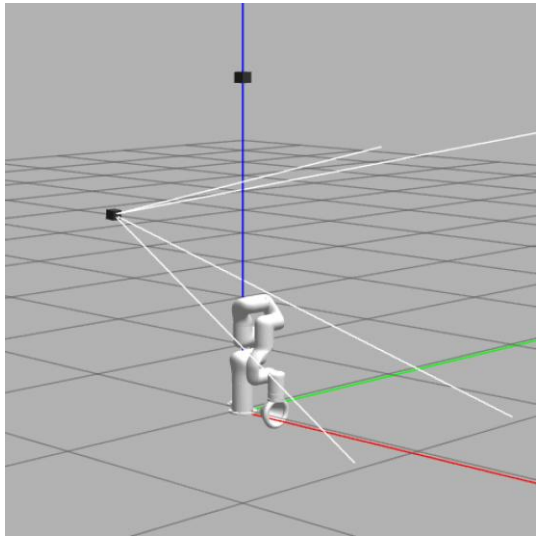
- **Catches Ball**
  - Tennis ball
  - Underhand throw
  - 3 m/s groundspeed
  - 90% Catch success (50% with obstacles)
- **Avoids Obstacles**
  - Cylinders 30mm dia. x 500 mm len
  - Only in outer 50% of Catch Volume
  - >10% trials include a collision
- **Environment**
  - Indoors (favorable lighting)
  - 3m - 5m throwing range
  - 25° max heading from camera center
  - Forward half of reach (↓ from full volume)
- **Other**
  - 3.14 rads/s max joint speed
  - 20 rads/s^2 max joint accel
  - <$8700 total project cost

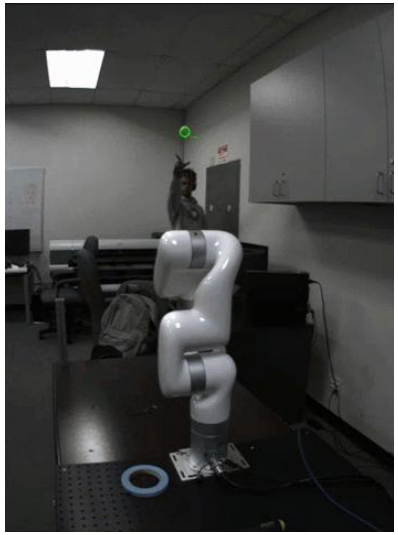


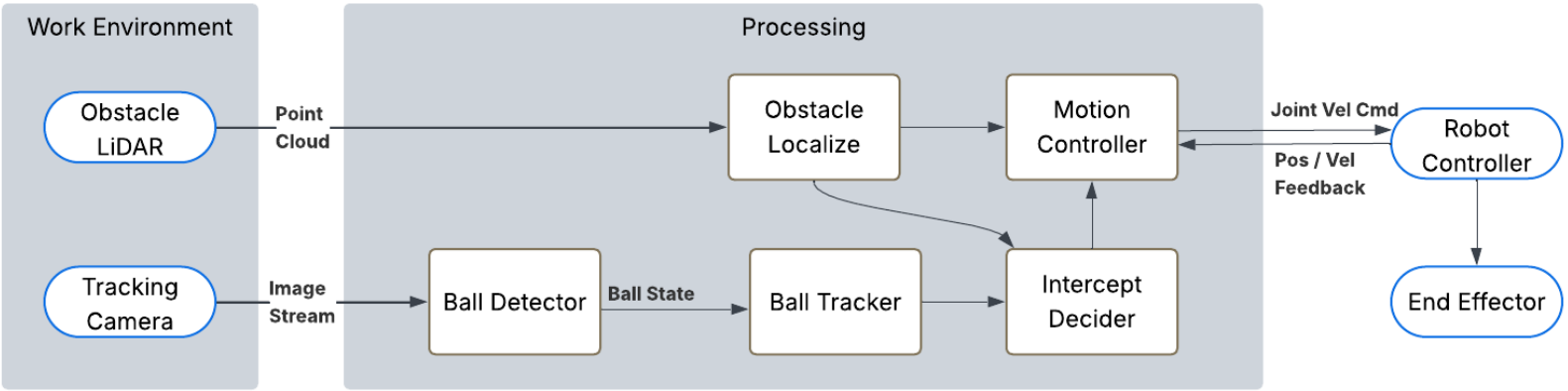


System Design









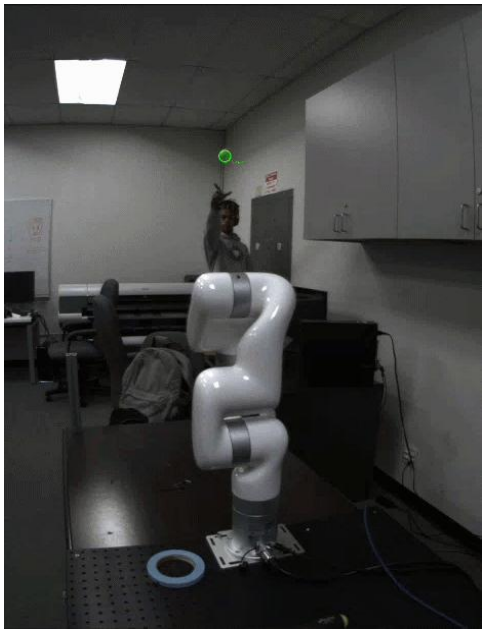
IEEE 1872.2 – Autonomous Robotics Ontologies [4]

Project Cost

Cost Table

Item	Cost ($)	Need to Purchase?
Robotic Arm	$6000	Purchased
LiDAR Depth Camera	$0	No
Tracking Camera (FLIR Blackfly)	$370	Purchased
Camera Lens	$144	Purchased
Camera Accessories	$136	Purchased
Testing Equipment	$400 - 700	Yes
Total Cost	$7050 - $7350 (↓ $7800)	

Ball Tracking / Intercept

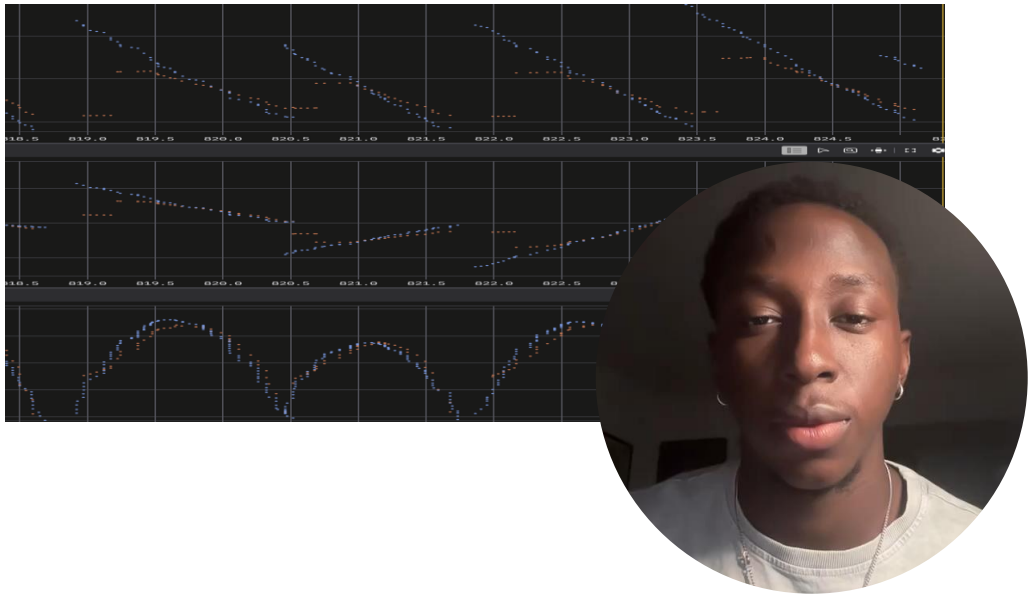


Current Progress

- Blackfly S camera integrated with ball_detector_node, successfully tracking ball in real hardware environment using HSV color filtering
- Trajectory predictor (ball_trajectory_predictor_node) mapping estimated ball arc using projectile kinematics at 0.05s intervals
- 6-state Kalman filter [x,y,z,vx,vy,vz] with gravity model running on live camera feed
- Monocular depth estimation validated from apparent ball radius

Next Steps

- Integrate trajectory prediction output with intercept_decider_node for closed-loop testing on xArm5
- Test with entire system



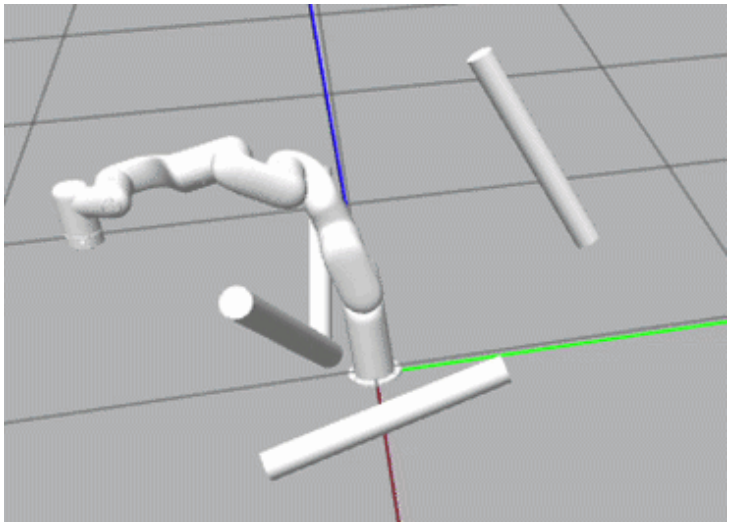
Obstacle Avoidance

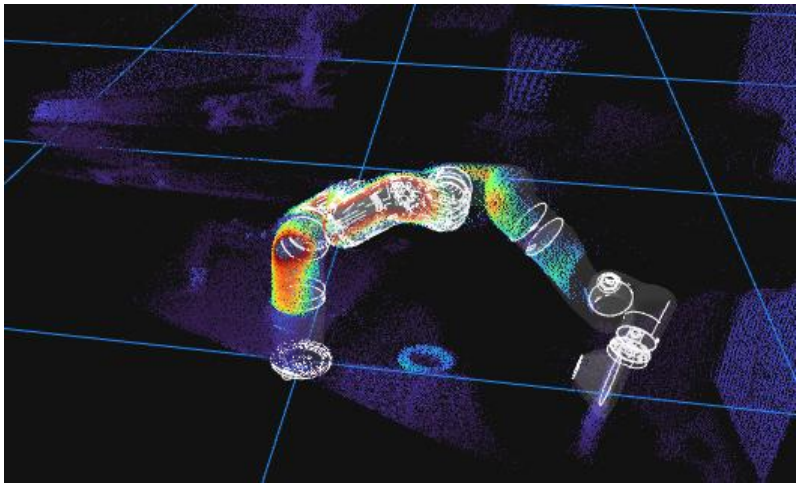
Current Progress

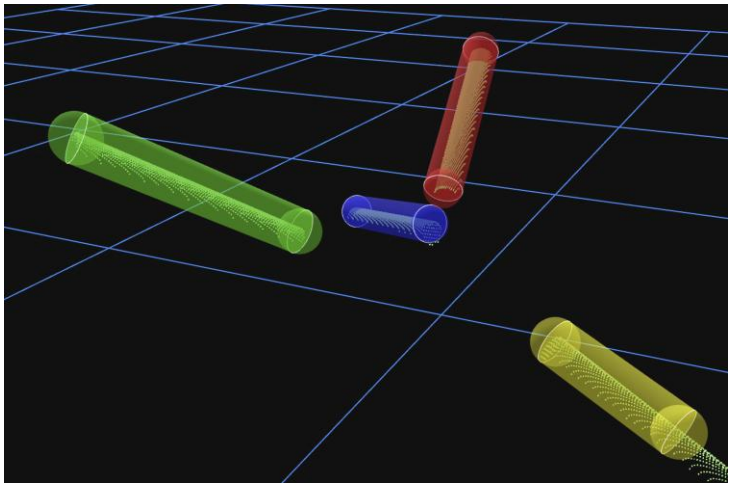
- Helios 2+ Integrated and calibrated
- ROS2 node integrated and tested
- Point cloud clustering
- Obstacle avoidance constraints on MPC

Next Steps

- Real obstacle detection and avoidance
- Tune and improve efficiency







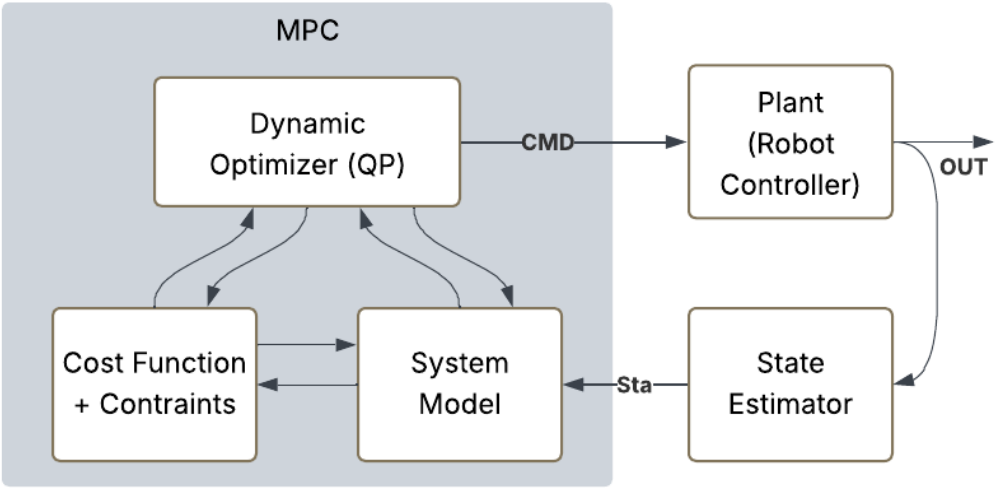
Motion Control / MPC

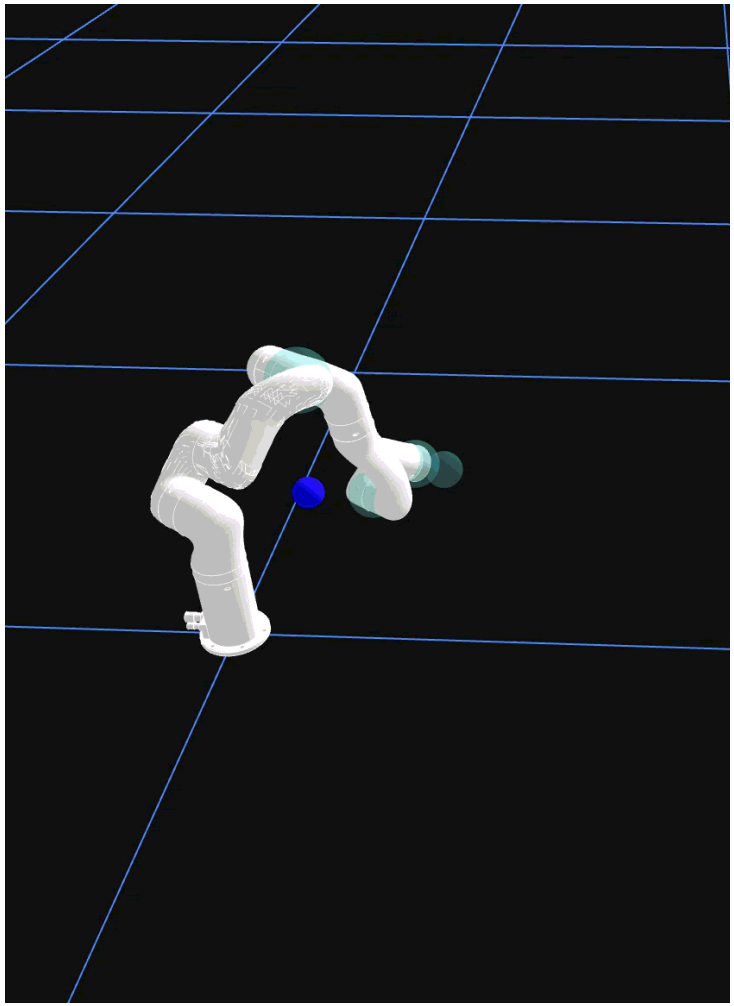
Current Subsystem Progress

- Implemented advanced Model Predictive Control for fast intercept and obstacle avoidance
- Added collision avoidance and reset safety

Next Steps

- Improve safety constraints and responsiveness





IEEE 1228 – Software Safety Plans [2]

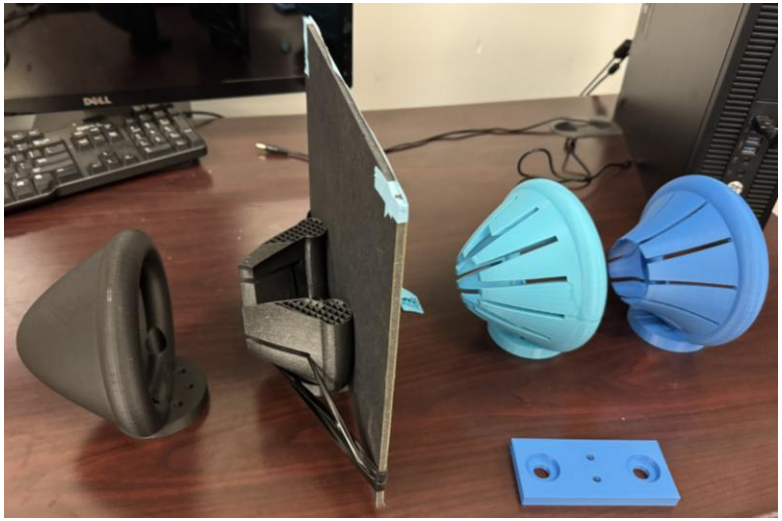
End Effector

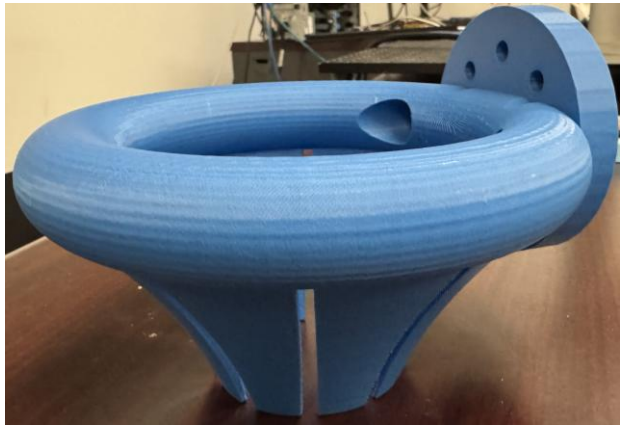
Current Subsystem Progress

- Five end effector designs have been modeled and printed
- The fourth iteration has been mounted
- Durability and catchability testing is progressing well

Next Steps

- Look into new end effector designs
- Establish controlled quantifiable testing to compare the accuracies of the different iterations made

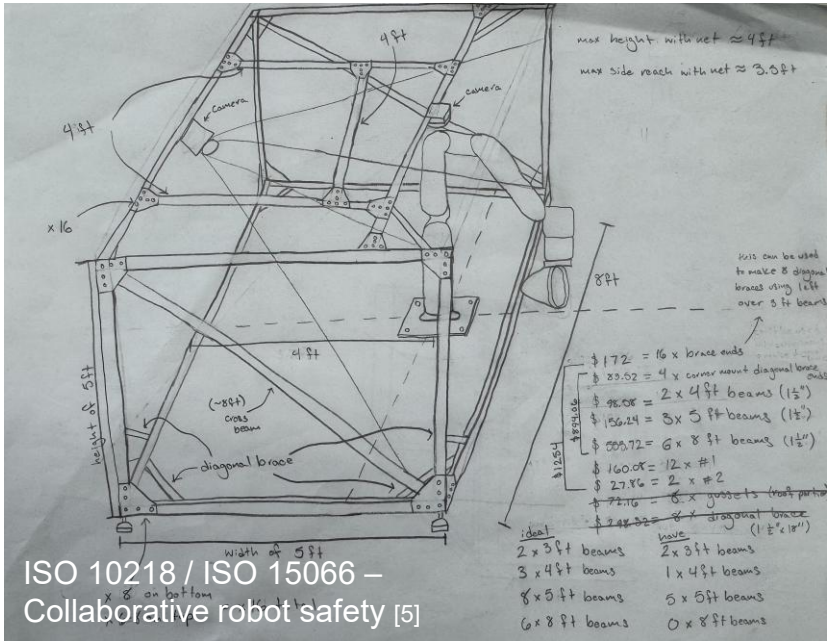




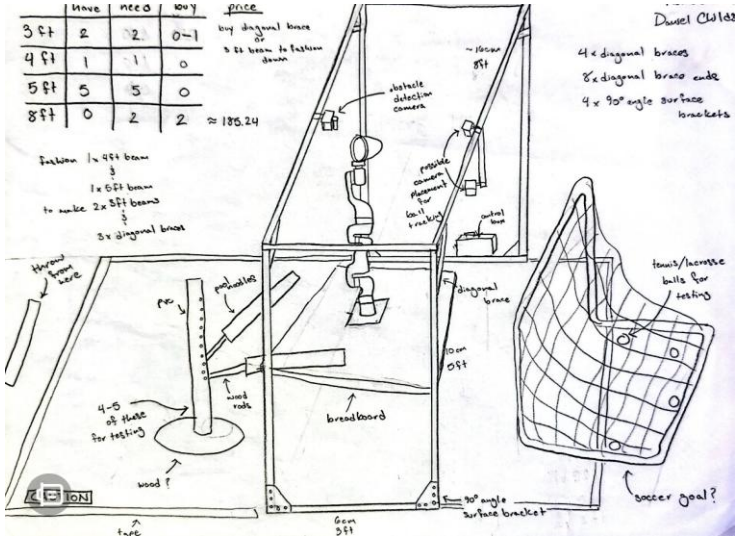
Robot Work Environment

Current Subsystem Progress

- Six work environment designs have been drawn out and itemized with detailed inventories attached
- The sixth iteration has been submitted for order
- 3D printing is in progress to adapt the cameras to our work environment

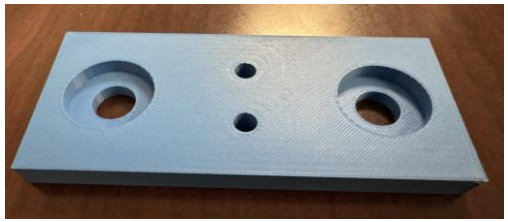


ISO 10218 / ISO 15066 – Collaborative robot safety [5]



Next Steps

- Look into new minimal designs for the work environment, camera mounts, and robot obstacles
- Construct the final work environment design



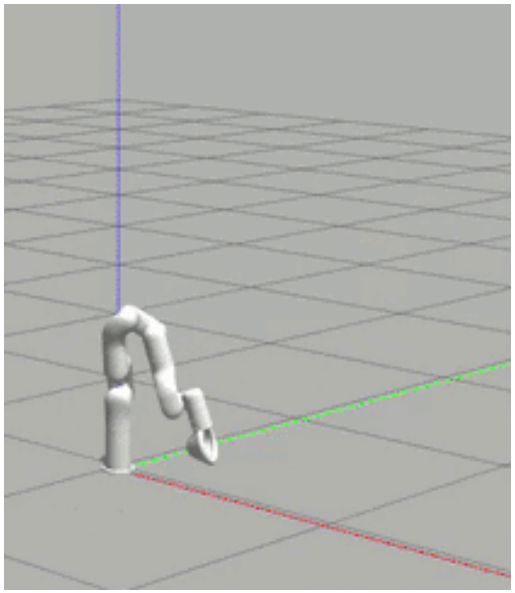
Integration

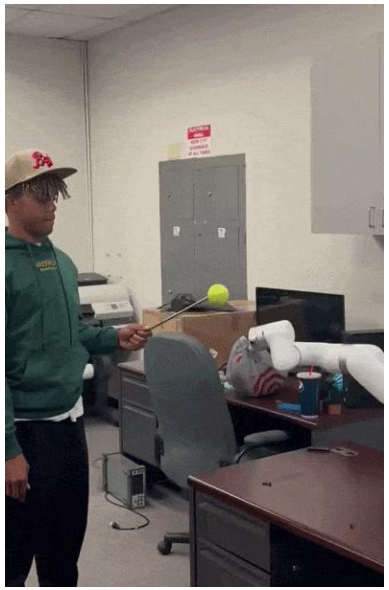
Current Subsystem Progress

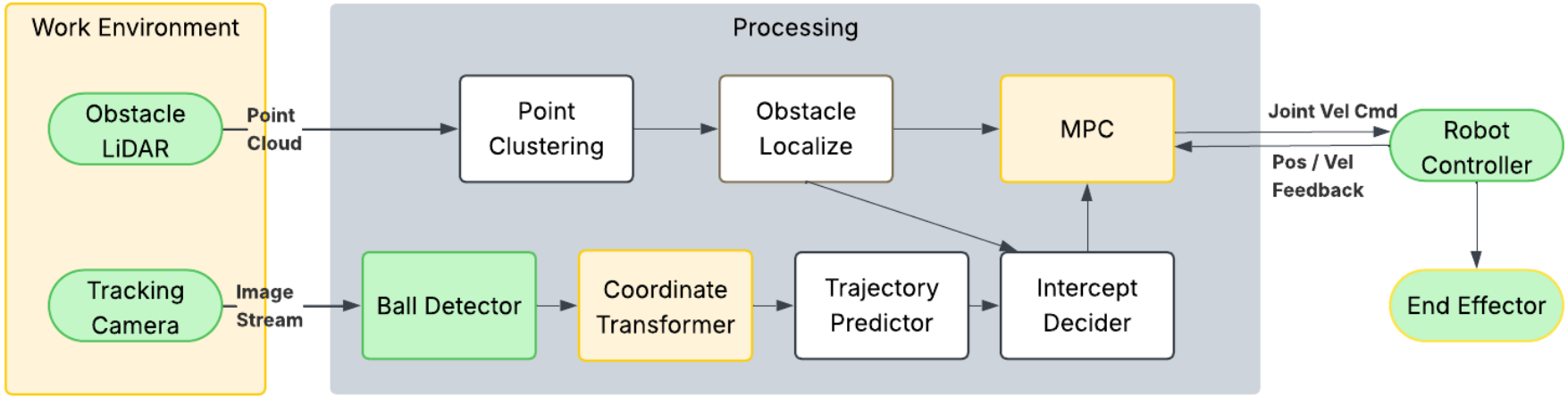
- Sim catching fully integrated, 85 – 90% catch rate (ground truth ball state, no obstacles)
- Hardware fully integrated, first tests unsuccessful

Next Steps

- Explore solutions for motion planning







Milestones / Timeline

Prototyping	ROS Integration	HW Integration	Testing	Final Demo
Oct-Dec '25	Jan-Mar '26	Jan-Mar '26	Mar-Apr '26	May '26

Completed

✓ Build ROS node/topic structure	*Jan 25*
✓ Create Sim testing environment	*Jan 29*
✓ Ball Tracker Node	*Feb 5*
✓ MPC Node	*Feb 9*
✓ Configure & calibrate camera	*Feb 19*
✓ Intercept Decider / Trajectory Predictor	*Feb 21*
✓ Obstacle Localizer	*Mar 6*

In Progress & Next Up

LiDAR Integration	95%
Ball Tracker Live Integration	85%
Sensor Mounting Frame	75%
Live Arm Motion Planning	70%
Final End Effector	60%
Refine Algorithms / Integration	30%
Final Acceptance Testing	0%

IEEE 1028 – Software Reviews and Audits [3]

Thank You!

Sources

[1] United Nations, “THE 17 GOALS,” Sustainable Development Goals. [Online]. Available: https://sdgs.un.org/goals. Accessed: Oct. 27, 2025. (Sustainable Development Goals)

[2] IEEE, *IEEE Std 1228-1994: IEEE Standard for Software Safety Plans*. New York, NY, USA: IEEE, 1994. (Antpedia)

[4] IEEE, IEEE Std 1872.2-2021: IEEE Standard Ontologies for Robotics and Automation—Part 2: Autonomous Robotics (AuR) Ontology. New York, NY, USA: IEEE, 2021. (IEEE Xplore)

[3] IEEE, *IEEE Std 1028-2008: IEEE Standard for Software Reviews and Audits*. New York, NY, USA: IEEE, 2008. (ANSI Webstore)

[5] ISO, *ISO 10218-1:2025—Robotics—Safety requirements—Part 1: Industrial robots*. Geneva, Switzerland: ISO, 2025. (ISO)

[5] ISO, *ISO 10218-2:2025—Robotics—Safety requirements—Part 2: Industrial robot applications and robot cells*. Geneva, Switzerland: ISO, 2025. (ISO)

[6] IEC, IEC 61508:2010—Functional safety of electrical/electronic/programmable electronic safety-related systems. Geneva, Switzerland: International Electrotechnical Commission (IEC), 2010. (IEC)